mathematician Takakazu Seki Kowa wrote a book called "Method of Solving the Dissimulated Problems." This book gives general methods for calculating determinants and presents examples for matrices as large as 5×5. Coincidentally, in the same year (1683) Gottfried Leibniz in Europe also first used determinants to solve systems of linear equations. Leibniz also discovered that a determinant could be expanded using any of the matrix columns.

In the middle of the 1700s, Colin Maclaurin and Gabriel Cramer published some major contributions to matrix theory. After that point, work on matrices became rather regular, with significant contributions by Etienne Bezout, Alexandre Vandermonde, Pierre Laplace, Joseph Lagrange, and Carl Gauss. The term "determinant" was first used in the modern sense by Augustin Cauchy in 1812 (although the word was used earlier by Gauss in a different sense). Cauchy also discovered matrix eigenvalues and diagonalization, and introduced the idea of similar matrices. He was the first to prove that every real symmetric matrix is diagonalizable.

James Sylvester (in 1850) was the first to use the term "matrix." Sylvester moved to England in 1851 to became a lawyer and met Arthur Cayley, a fellow lawyer who was also interested in mathematics. Cayley saw the importance of the idea of matrices and in 1853 he invented matrix inversion. Cayley also proved that 2×2 and 3×3 matrices satisfy their own characteristic equations. The fact that a matrix satisfies its own characteristic equation is now called the Cayley–Hamilton theorem (see Problem 1.5). The theorem has William Hamilton's name associated with it because he proved the theorem for 4×4 matrices during the course of his work on quaternions.

Camille Jordan invented the Jordan canonical form of a matrix in 1870. Georg Frobenius proved in 1878 that all matrices satisfy their own characteristic equation (the Cayley Hamilton theorem). He also introduced the definition of the rank of a matrix. The nullity of a square matrix was defined by Sylvester in 1884. Karl Weierstrass's and Leopold Kronecker's publications in 1903 were instrumental in establishing matrix theory as an important branch of mathematics. Leon Mirsky's book in 1955 [Mir90] helped solidify matrix theory as a fundamentally important topic in university mathematics.

1.2 LINEAR SYSTEMS

Many processes in our world can be described by state-space systems. These include processes in engineering, economics, physics, chemistry, biology, and many other areas. If we can derive a mathematical model for a process, then we can use the tools of mathematics to control the process and obtain information about the process. This is why state-space systems are so important to engineers. If we know the state of a system at the present time, and we know all of the present and future inputs, then we can deduce the values of all future outputs of the system.

State-space models can be generally divided into linear models and nonlinear models. Although most real processes are nonlinear, the mathematical tools that are available for estimation and control are much more accessible and well understood for linear systems. That is why nonlinear systems are often approximated as linear systems. That way we can use the tools that have been developed for linear systems to derive estimation or control algorithms.

A continuous-time, deterministic linear system can be described by the equations

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned} \tag{1.67}$$

where x is the state vector, u is the control vector, and y is the output vector. Matrices A, B, and C are appropriately dimensioned matrices. The A matrix is often called the system matrix, B is often called the input matrix, and C is often called the output matrix. In general, A, B, and C can be time-varying matrices and the system will still be linear. If A, B, and C are constant then the solution to Equation (1.67) is given by

$$\begin{aligned} x(t) &= e^{A(t-t_0)}x(t_0) + \int_{t_0}^{t} e^{A(t-\tau)}Bu(\tau)\,d\tau \\ y(t) &= Cx(t) \end{aligned} \tag{1.68}$$

where t_0 is the initial time of the system and is often taken to be 0. This is easy to verify when all of the quantities in Equation (1.67) are scalar, but it happens to be true in the vector case also. Note that in the zero input case, $x(t)$ is given as

$$x(t) = e^{A(t-t_0)}x(t_0), \qquad \text{zero input case} \tag{1.69}$$

For this reason, e^{At} is called the state-transition matrix of the system.[3] It is the matrix that describes how the state changes from its initial condition in the absence of external inputs. We can evaluate the above equation at $t = t_0$ to see that

$$e^{A0} = I \tag{1.70}$$

in analogy with the scalar exponential of zero.

As stated above, even if x is an n-element vector, then Equation (1.68) still describes the solution of Equation (1.67). However, a fundamental question arises in this case: How can we take the exponential of the matrix A in Equation (1.68)? What does it mean to raise the scalar e to the power of a matrix? There are many different ways to compute this quantity [Mol03]. Three of the most useful are the following:

$$\begin{aligned} e^{At} &= \sum_{j=0}^{\infty} \frac{(At)^j}{j!} \\ &= \mathcal{L}^{-1}[(sI - A)^{-1}] \\ &= Qe^{\hat{A}t}Q^{-1} \end{aligned} \tag{1.71}$$

The first expression above is the definition of e^{At}, and is analogous to the definition of the exponential of a scalar. This definition shows that A must be square in order for e^{At} to exist. From Equation (1.67), we see that a system matrix is always square. The definition of e^{At} can also be used to derive the following properties.

$$\begin{aligned} \frac{d}{dt}e^{At} &= Ae^{At} \\ &= e^{At}A \end{aligned} \tag{1.72}$$

[3]The MATLAB function EXPM computes the matrix exponential. Note that the MATLAB function EXP computes the element-by-element exponential of a matrix, which is generally not the same as the matrix exponential.

In general, matrices do not commute under multiplication but, interestingly, a matrix always commutes with its exponential.

The first expression in Equation (1.71) is not usually practical for computational purposes since it is an infinite sum (although the latter terms in the sum often decrease rapidly in magnitude, and may even become zero). The second expression in Equation (1.71) uses the inverse Laplace transform to compute e^{At}. In the third expression of Equation (1.71), Q is a matrix whose columns comprise the eigenvectors of A, and $\hat{A}$ is the Jordan form[4] of A. Note that Q and $\hat{A}$ are well defined for any square matrix A, so the matrix exponential e^{At} exists for all square matrices A and all finite t. The matrix $\hat{A}$ is often diagonal, in which case $e^{\hat{A}t}$ is easy to compute:

$$\begin{aligned}
\hat{A} &= \begin{bmatrix} \hat{A}_{11} & 0 & \cdots & 0 \\ 0 & \hat{A}_{22} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & \hat{A}_{nn} \end{bmatrix} \\
e^{\hat{A}t} &= \begin{bmatrix} e^{\hat{A}_{11}} & 0 & \cdots & 0 \\ 0 & e^{\hat{A}_{22}} & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \cdots & e^{\hat{A}_{nn}} \end{bmatrix}
\end{aligned} \tag{1.73}$$

This can be computed from the definition of $e^{\hat{A}t}$ in Equation (1.71). Even if the Jordan form matrix $\hat{A}$ is not diagonal, $e^{\hat{A}t}$ is easy to compute [Bay99, Che99, Kai80]. We can also note from the third expression in Equation (1.71) that

$$\begin{aligned}
\left[e^{At}\right]^{-1} &= e^{-At} \\
&= Qe^{-\hat{A}t}Q^{-1}
\end{aligned} \tag{1.74}$$

(Recall that A and $-A$ have the same eigenvectors, and their eigenvalues are negatives of each other. See Problem 1.10.) We see from this that e^{At} is always invertible. This is analogous to the scalar situation in which the exponential of a scalar is always nonzero.

Another interesting fact about the matrix exponential is that all of the individual elements of the matrix exponential e^A are nonnegative if and only if all of the individual elements of A are nonnegative [Bel60, Bel80].

■ EXAMPLE 1.2

As an example of a linear system, suppose that we are controlling the angular acceleration of a motor (for example, with some applied voltage across the motor windings). The derivative of the position is the velocity. A simplified motor model can then be written as

[4] In fact, Equation (1.71) can be used to define the Jordan form of a matrix. That is, if e^{At} can be written as shown in Equation (1.71), where Q is a matrix whose columns comprise the eigenvectors of A, then $\hat{A}$ is the Jordan form of A. More discussion about Jordan forms and their computation can be found in most linear systems books [Kai80, Bay99, Che99].

$$\begin{aligned}\dot{\theta} &= \omega \\ \dot{\omega} &= u + w_1 \end{aligned} \tag{1.75}$$

The scalar w_1 is the acceleration noise and could consist of such factors as uncertainty in the applied acceleration, motor shaft eccentricity, and load disturbances. If our measurement consists of the angular position of the motor then a state space description of this system can be written as

$$\begin{aligned}\begin{bmatrix} \dot{\theta} \\ \dot{\omega} \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \theta \\ \omega \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 0 \\ w_1 \end{bmatrix} \\ y &= \begin{bmatrix} 1 & 0 \end{bmatrix} x + v \end{aligned} \tag{1.76}$$

The scalar v consists of measurement noise. Comparing with Equation (1.67), we see that the state vector x is a 2×1 vector containing the scalars θ and ω.

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■ EXAMPLE 1.3

In this example, we will use the three expressions in Equation (1.71) to compute the state-transition matrix of the system described in Example 1.2. From the first expression in Equation (1.71) we obtain

$$\begin{aligned} e^{At} &= \sum_{j=0}^{\infty} \frac{(At)^j}{j!} \\ &= (At)^0 + (At)^1 + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots \\ &= I + At \end{aligned} \tag{1.77}$$

where the last equality comes from the fact that $A^k = 0$ when $k > 1$ for the A matrix given in Example 1.2. We therefore obtain

$$\begin{aligned} e^{At} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & t \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \end{aligned} \tag{1.78}$$

From the second expression in Equation (1.71) we obtain

$$\begin{aligned} e^{At} &= \mathcal{L}^{-1}[(sI - A)^{-1}] \\ &= \mathcal{L}^{-1}\left(\begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix}^{-1}\right) \\ &= \mathcal{L}^{-1}\begin{bmatrix} 1/s & 1/s^2 \\ 0 & 1/s \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \end{aligned} \tag{1.79}$$